

Feudal hierarchies and reinforcement learning

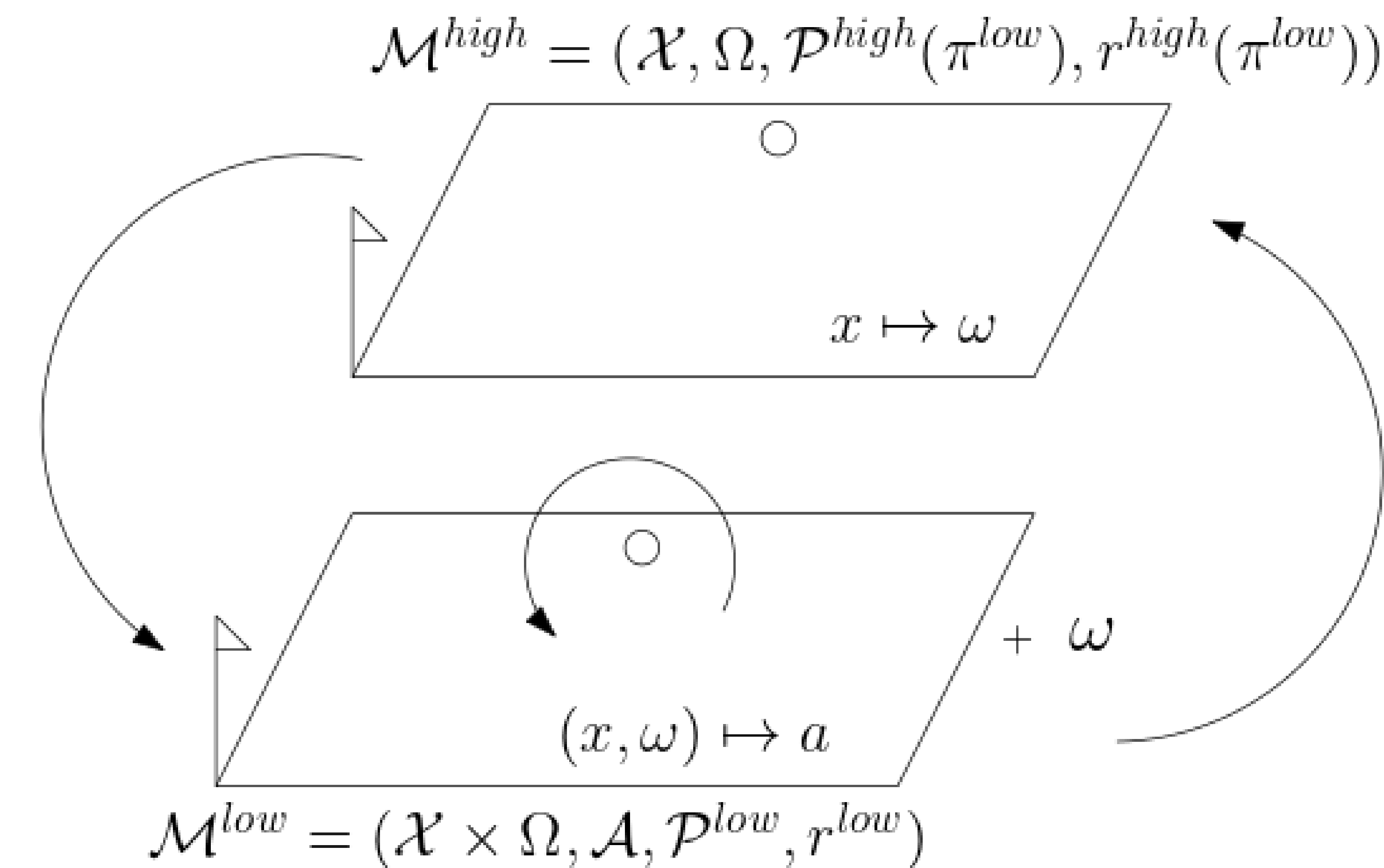
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Feudal hierarchies

High-level policy → learns to set goals.

Low-level policy → learn to achieve the goals.

What is the best goal to set **depends** on the low-level policy.



State space $\mathcal{X}$

Goal space Ω

$\pi^{high} : \mathcal{X} \rightarrow \Omega$

Action space $\mathcal{A}$

$\pi^{low} : \mathcal{X} \times \Omega \rightarrow \mathcal{A}$

Introduction

Hierarchical decision making induces a form of **non-stationarity** in an environment.

We show that feudal hierarchies are stable and reinforcement learning **converges**.

Results

1. In a feudal hierarchy, the low-level is a Markov decision process and the high-level is a Markov decision process for **each** low-level policy.
2. **Despite** the non-stationarity, while the low-level policy converges, the high-level Markov decision process also converges.
3. Reinforcement learning converges to the optimal solution on **both** levels.

Conclusion

We provide theoretical guarantees for the use of reinforcement learning in hierarchical decision making processes.

References

Peter Dayan and Geoffrey E Hinton. Feudal reinforcement learning. In S. Hanson, J. Cowan, and C. Giles, editors, Advances in Neural Information Processing Systems, volume 5. Morgan-Kaufmann, 1993.

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